1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights immune/inflammation axes (EDIH, CSF-1/SPC), genomic instability (CENPA), and therapy-linked immune readouts (Olaparib → immune infiltrate). The key puzzle is: why do tumors with apparent in-situ immune organization (imaging-detectable TLS/TIL features) not uniformly benefit from therapies—specifically PARP inhibition—that may rely in part on immune-mediated mechanisms? 2) Tier 1 paths used: I built on the CSF-1 → macrophage/inflammation axis (CSF1/SPC) because it provides a clear mechanistic suppressive immune signal, and on CENPA because chromosomal instability / mitotic dysregulation can alter neoantigen burden and immune visibility. These two paths create a plausible interaction: tumor-driven immunosuppression (CSF1) + altered antigenic landscape (CENPA) that modulates the functional impact of organized lymphoid structures. 3) Tier 2 support: EDIH/inflammation signatures and metabolites support an inflammatory/metabolic microenvironment that can reinforce macrophage-mediated suppression; radiomics/pathomics and Decipher GC provide orthogonal, noninvasive ways to stratify tumors and could serve as validation axes for the interaction; PCA3 and SPC are ancillary readouts for tumor activity and CSF1-driven signaling respectively. 4) Endpoint choice: I selected therapy_response (PARP inhibitor response proxy) because the Ontologist explicitly linked Olaparib/PARP inhibition to immune infiltrate readouts, making a therapy-response question both mechanistically tractable and clinically actionable. 5) Deliberate exclusions: I excluded PCA3, Decipher GC, AFP and broad metabolite panels from the core hypothesis to keep the mechanistic question sharp and testable (max 2 anchors). Those entities are retained as secondary_modifiers for follow-up stratification or validation rather than core causal drivers in the hypothesis?

Working Hypothesis

TLS-like lymphoid aggregates indicate a latent anti-tumor immune potential, but their ability to mediate PARP inhibitor (PARPi)–driven tumor control is contingent on (i) the macrophage-influenced immunosuppressive milieu governed by CSF1/CSF1R signaling and (ii) the tumor's chromosomal instability profile as reflected by CENPA activity, which shapes neoantigen quality and immune visibility. Specifically, TLS-high tumors with low CSF1 signaling and low CIN/CENPA activity will exhibit robust PARPi-induced anti-tumor immunity and clinical response. In contrast, TLS-high tumors with high CSF1 expression and/or high CIN/CENPA activity will show blunted PARPi responses due to macrophage-mediated suppression and suboptimal neoantigen presentation. Consequently, two actionable refinements emerge: (1) combining PARPi with CSF1R blockade or macrophage-repolarizing therapies should improve responses in TLS-high, CSF1-high/CIN-high tumors, and (2) stratification by TLS density plus CSF1 and CIN/CENPA status will identify a subset of patients most likely to benefit from PARPi monotherapy or combination therapy. This framework yields testable predictions across in vitro, in vivo, and patient-derived models and can be validated with noninvasive radiomics/pathomics readouts as orthogonal stratifiers.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CSF1, CENPA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CSF1: CSF1/SPC) because it provides a clear mechanistic suppressive immune signal, and on CENPA because chromosomal instability / mitotic dysregulation can alter neoantigen burden and immune visibility.

CENPA: CENPA because chromosomal instability / mitotic dysregulation can alter neoantigen burden and immune visibility.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EDIH	Supporting modifier from knowledge graph
Secondary	PCA3	Supporting modifier from knowledge graph
Secondary	Decipher_GC	Supporting modifier from knowledge graph
Secondary	SPC	Supporting modifier from knowledge graph
Secondary	metabolites_biomarkers	Supporting modifier from knowledge graph
Secondary	radiomics_features	Supporting modifier from knowledge graph
Secondary	pathomics_features	Supporting modifier from knowledge graph
Secondary	AFP (exploratory)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In vitro co-cultures of BRCA1/2-deficient cancer cell lines with human monocyte-derived macrophages; manipulate CSF1/CSF1R signaling (add/remove CSF1, CSF1R inhibitors) and treat with olaparib; readouts: macrophage polarization (CD206, iNOS), cytokine milieu (IL-10, TGF-β, IL-

Prediction 2: , T cell activation markers in co-cultures, and cancer cell viability.

Prediction 3: In vivo murine models: BRCA-deficient tumor models with controlled CENPA expression (CRISPR/Cas9 knockdown or overexpression to modulate CIN) and with or without CSF1R blockade; administer olaparib ± CSF1R inhibitor; endpoints include tumor growth, survival, TLS formation by multiplex IHC, macrophage phenotype, and T cell infiltration.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EDIH: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PCA3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.